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Trends in Genetics

Review

Gene expression noise in development: genome-wide dynamics

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Gene expression noise underlies cell-to-cell variability in RNA and protein levels of a seemingly homogeneous population of cells. Emerging evidence suggests a functional role for this variability in the specification of cell fates during mammalian development. Advances in genome-wide and single-cell technologies now enable the quantification and deciphering of transcriptome variability with increasing precision. In this review, we highlight recent insights into the significance of gene expression noise during early embryogenesis, focusing on RNA variability. We discuss new approaches to further quantify and unravel different sources of gene expression noise and how this yields insights into early mammalian development.

Highlights

Novel methodological approaches are starting to decipher and quantify gene expression noise at a genome-wide scale.

Cell-to-cell RNA variability increases towards a cellular lineage bifurcation and declines after cell fate commitment.

Manipulation of gene expression noise introduces cell fate bias or affects cell-state-switching dynamics.

During early embryonic development, pluripotency and development genes consistently demonstrate high intrinsic noise, primarily caused by transcriptional bursting.

Promoter structure and epigenomic make-up are globally correlated to bursting parameters of genes.

Cobursting is a rare event and is mostly restricted to either paralogues or genes in close spatial proximity.

Transcriptional bursting and RNA degradation seem to be synergistic and can lead to increased or decreased noise for specific classes of genes.

Gene expression noise in a developmental context

Cellular noise (see Glossary) refers to the phenotypic variability observed in genetically identical cells from the same cell type, when they share the same microenvironment. One of the main contributors to cellular noise is **gene expression noise**, arising from random fluctuations in RNA and protein levels and leading to cell-to-cell variability (Figure 1). To better understand gene expression noise, early studies have separated intrinsic and extrinsic sources [2,3]. **Extrinsic noise** results from differences between cells, such as distinct cell cycle stages, or variability in the levels of regulatory molecules that affect the entire cell in a global manner. Conversely, **intrinsic noise** originates from the inherent stochasticity of biochemical processes within a cell, particularly those involving low-copy numbers, such as transcription, often referred to as **transcriptional noise** [4]. Pioneering studies in mammals visualising mRNA at the single-molecule level revealed that mRNA production occurs in bursts, the major contributor to transcriptional noise [5]. However, post-transcriptional processes, such as RNA processing, nuclear export, translation, and protein and **RNA degradation**, also affect cell-to-cell variability in RNA and protein levels [6]. Currently, genome-wide gene expression levels are predominantly measured at the single-cell level using RNA sequencing (scRNA-seq). In this review, we therefore focus on advanced RNA profiling approaches and use the term 'gene expression noise' to refer to the variability in RNA levels between individual cells, unless otherwise specified.

As a result of fluctuating gene expression levels, gene expression noise has the ability to influence cellular states [7–9]. As such, it is associated with dynamic systems that require cellular response, adaptation, or change. For mammalian cell fate decisions, prominently studied using mouse embryonic stem cells (mESCs), the variability in gene expression is increasingly being characterised [10]. In nonmammalian systems, dedicated approaches have started to appreciate the functional role of gene expression noise (Box 1) [16,19], but understanding this functionality in mammals remains challenging. This raises a fundamental question in developmental biology: is the high level of gene expression noise, as observed in mESCs, functional during early embryogenesis? Exciting developments in scRNA-seq approaches make it possible to assess variability at a transcriptome-wide scale [25]. This sparked further interest: can genome-wide approaches

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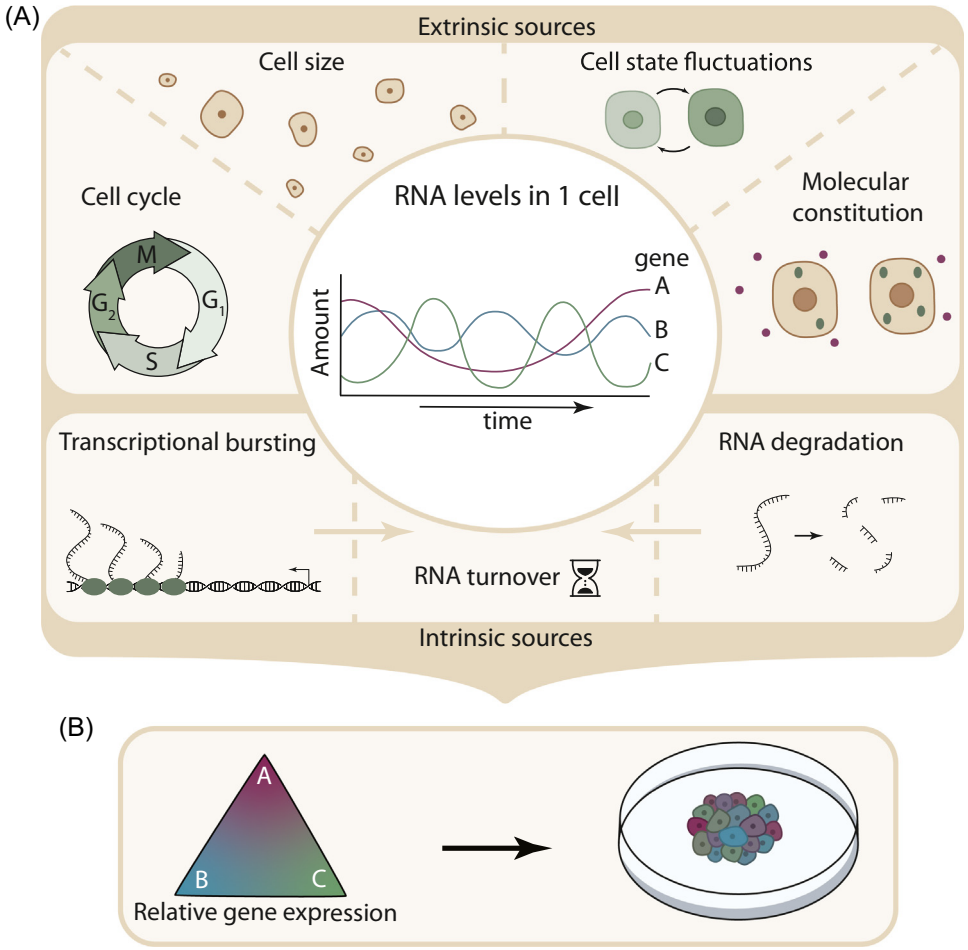


Figure 1. Sources of gene expression noise. (A) Main cellular features associated with gene expression noise (depicted for three genes in the middle panel). Here, we classified each feature into the two defined sources, extrinsic noise (top panel) and intrinsic noise (bottom panel), but it should be noted that the events associated here with intrinsic noise are also subject to extrinsic noise [1]. Extrinsic noise sources include, but are not limited to, cell cycle states, cell size, cell state fluctuations, and level of molecular components within the cell (such as RNA polymerases) and outside of the cell (such as signalling molecules). The two main components of intrinsic noise are transcriptional bursting and RNA degradation, which together influence RNA turnover. As RNA synthesis and degradation rates differ across genes, change over time, and are asynchronous within a cell population, this results in gene expression noise, i.e., fluctuating RNA levels and variability between cells (middle panel). (B) Gene expression noise results in cellular heterogeneity, as visualised here by the relative expression levels of the three genes depicted in the middle panel of A, shown in a clonal cell population.

that measure RNA synthesis and degradation parameters reveal how gene expression noise propagates throughout the mRNA life cycle [26,27]? In this review, we highlight recent discoveries that shed light on the degree and potential roles of gene expression variability in mammalian development, including novel approaches to experimentally manipulate and **decipher gene expression noise**.

Significance of gene expression noise during mammalian development

Early studies have shown that important developmental regulators, such as *Nanog* [7,8], *Hes1* [28], and *Rex1* [29], show variable protein levels between individual mESCs. More recently, scRNA-seq allowed for the measurement of gene expression noise at a genome-wide level

Glossary

Cellular noise: the phenotypic variability in molecular content between genetically identical cells of the same type in identical environments, such as variability in abundance of protein and RNA.
Coboursting: the event in which transcriptional bursts for two or more genes or for the two alleles of one gene (allelic coboursting) occur simultaneously more often than expected by random chance. It is also referred to as coordinated bursting.
Correlated transcription: simultaneous RNA production between two or more genes. This does not necessarily indicate simultaneous transcriptional burst initiation.
Decipher gene expression noise: the breakdown of gene expression noise on RNA level by quantifying its underlying sources, such as transcriptional bursting, RNA degradation, and cell cycle stage.
Extrinsic noise: variability in gene expression of a homogeneous cell population resulting from differences in cellular or external factors. Cell cycle stage, cell size, fluctuating cell states (such as Nanog high/low), or concentrations of external (signalling) or internal (RNA polymerases) molecules are all examples of extrinsic noise.
Gene expression noise: variability in gene expression of a homogeneous isogenic cell population.
***In vitro*:** a highly controlled environment in which cells can be grown outside of a living organism.
***In vivo*:** the natural environment of cells in a living organism.
Intrinsic noise: one of the contributors to gene expression noise. It arises from the inherent stochastic biochemical processes involving low concentrations of molecules within a cell, such as the stochastic nature in which RNA polymerases initiate gene transcription.
RNA degradation: the process of degrading a single RNA molecule, which can be quantified by degradation rate or RNA half-life.
RNA half-life: the time after which half of the gene-specific RNA molecules are degraded.
RNA stability: a measure of how long an RNA for a specific gene, on average, remains functionally present in the cell.
RNA turnover: the dynamic process by which RNA molecules are synthesised and degraded, determining how rapidly the RNA pool is replenished.

Box 1. The significance of gene expression noise in nonmammalian development

As studying the role of gene expression noise during *in vivo* mammalian embryonic development is challenging due to the inaccessibility of the embryos and relative complexity of mammalian embryogenesis, many of the current insights originate from nonmammalian systems. Embryos from *Drosophila melanogaster* (fruit fly) and *Danio rerio* (zebra fish) provide powerful model systems as these develop externally and rapidly divide [11,12]. In addition, zebra fish embryos are transparent [11], while fruit fly embryos have the advantage of forming a single embryonic layer [12]. This, for example, facilitates quantitative (live-cell) imaging to assay transcriptional bursting [13,14]. Moreover, external embryogenesis allows for convenient manipulation of processes associated with gene expression noise.

In the developing zebra fish, smFISH has shown that genes are stochastically activated during zygotic genome activation, leading to cytoplasmic cell-to-cell variability similar to mammals [15]. Interestingly, and seemingly different from mammals (Figure 2A), this variability decreases towards gastrulation, likely due to a buffering effect of increasing transcript density. The necessity of gene expression noise during zebra fish development is further demonstrated during the formation of hindbrain rhombomeres, which require intracellular fluctuations of *hoxb1a* and *krox20* for correct patterning [16].

Similarly, (sm)FISH studies show that gene expression occurs stochastically during fruit fly development, leading to noise [17,18]. During very early development, the noise between nuclei gets buffered by the syncytium, resulting in precise levels of mRNA in the shared cytoplasm of the nuclei [17]. Stochastic transcription seems to be essential during fruit fly embryogenesis as it contributes to developmental patterning [14]. It is also involved in eye development, where stochastic bursts of *spineless* are necessary for cell fate specification within retina cells [19], while specifically altering the bursting pattern of *spineless* leads to a different ratio of photoreceptor cell types [20]. More mechanistically, quantitative single-cell live imaging using stem-loops in fruit fly embryos showed coregulated transcriptional dynamics over large genomic distances [21]. This regulatory interconnectivity seemed to depend on promoter-proximal tethering elements and regulated bursting dynamics, highlighting a role for cotranscriptional hubs in transcription regulation and noise. Similarly, cobursting has been demonstrated in yeast, where neighbouring divergent and tandem *GAL* genes were temporally coupled [22].

A functional role for gene expression noise has also been shown for less complex organisms. Isogenic populations of viruses, bacteria, and yeast exploit variable gene expression to generate phenotypic diversity, facilitating cell state transitions and improving fitness and survival under environmental stress [23,24].

Transcriptional bursting: a phenomenon in which genes, rather than at a constant rate, get transcribed in a burst of activity followed by a period of inactivity. This bursting phenotype is due to transitions between active and inactive promoter states.
Transcriptional bursting parameters: parameters that allow the description of the transcriptional bursting kinetics, most commonly through burst frequency and burst size but also sometimes through burst initiation rate, burst amplitude, and burst duration.
Transcriptional noise: the noise arising from stochastic transcription and transcriptional bursting. See also intrinsic noise.

[30,31]. Early-stage mESCs exhibit high gene expression variability in cell cycle–associated genes, while later-stage mESCs show increased variability in developmental genes [30]. Live-cell imaging [32,33] and scRNA-seq [34], which allow the measurement of a gene's variability using the Fano factor (defined as the variance-to-mean ratio in expression) or more advanced metrics [35,36], have demonstrated that gene expression noise increases prior to lineage commitment, followed by a reduction in noise after commitment in ***in vitro*** cell cultures [32–35]. Reanalysis of multiple *in vitro* single-cell transcriptomic datasets during transitional states further revealed that cell-to-cell variability is higher immediately before cell fate decisions than in the starting cell population or after stable commitment, which the authors attribute to changes in bursting [37]. Together, these findings raise the questions: (i) to what extent do these results hold ***in vivo***, and (ii) does gene expression noise play a functional role at these stages?

Gene expression noise within mammalian embryos

Single-cell transcriptomics has enabled the comprehensive exploration of gene expression noise within mammalian embryos. Surprisingly, cell-to-cell variability has been detected within embryos as early as the two-cell stage in mice (Figure 2A) [38–40] and humans [39]. Transcriptional profiling of later stages shows that gene expression variability increases through the following four-cell (mouse) and eight-cell (human) stages [38–40]. Interestingly, a particularly strong rise in variability in mice and humans is observed during zygotic genome activation (ZGA) (Figure 2A). This suggests that active transcription is a major contributor to gene expression noise. Before ZGA, the observed variability is likely due to unequal cell cleavages [39,41].

From the mouse four-cell stage, the transcriptome-wide noise stabilises until the blastocyst stage [42]. Interestingly, at the blastocyst stage, cell-to-cell variability of the inner cell mass (ICM) seems to decline when the primitive endoderm and epiblast cell fates emerge [43]. During early

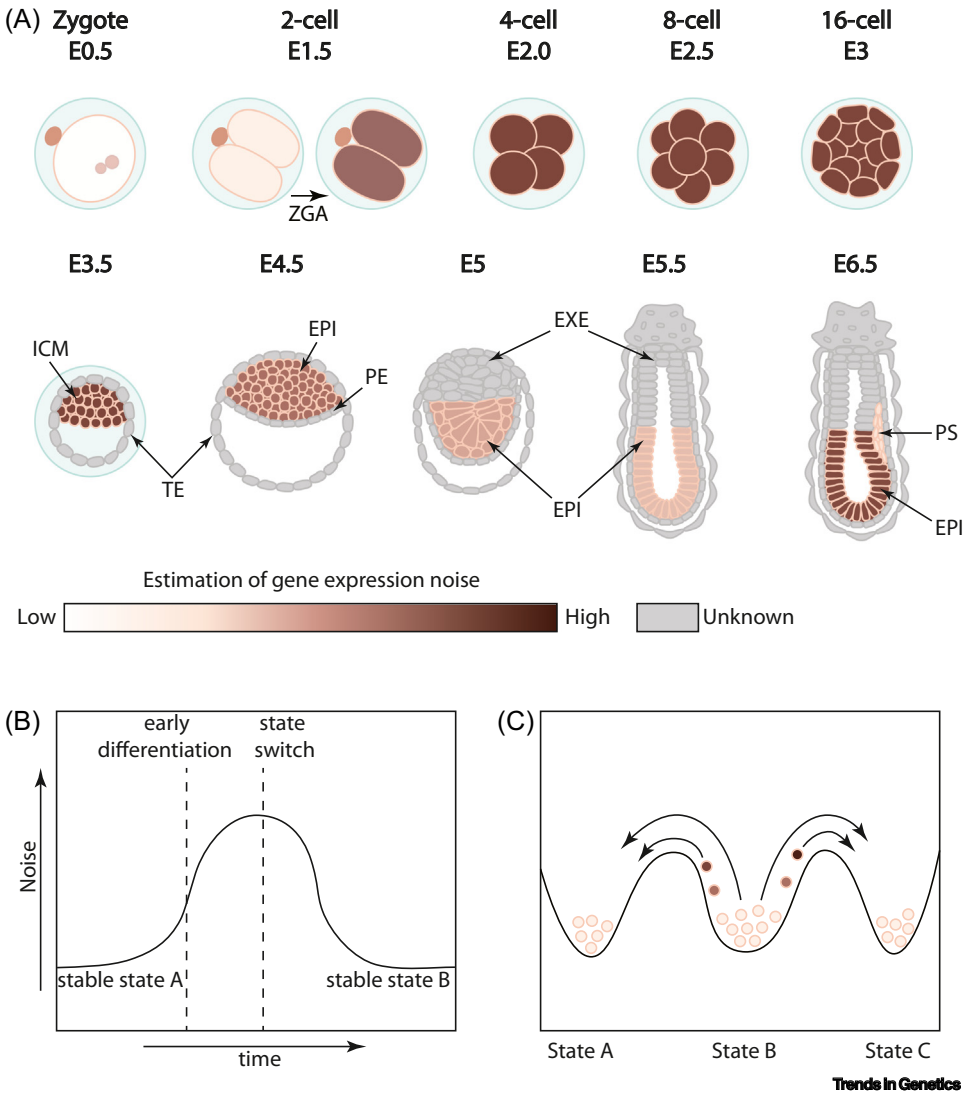


Figure 2. Gene expression noise during development. (A) Estimated level of cell-to-cell RNA variability during early mouse embryogenesis. Gene expression noise increases particularly during ZGA and decreases once cells become more committed, such as from the ICM to the EPI and from the EPI to the PS. (B) Gene expression noise increases as cells begin to differentiate and decreases once they switch states and stable expression patterns are established. (C) Increases in gene expression noise allow cells to explore different fates and lowers the barrier to switch cellular states. ZGA, zygotic genome activation; ICM, inner cell mass; EPI, epiblast; TE, trophectoderm; ExE, extraembryonic ectoderm; PS, primitive streak

gastrulation, RNA variability rises again in the epiblast at the onset of primitive streak differentiation, followed by a reduction once cells are committed [43] (Figure 2A). Thus, these *in vivo* findings reflect the *in vitro* studies in terms of gene expression noise during early fate transitions (Figure 2B). While this initial rise of cell-to-cell variability might be functional to facilitate cells exploring different fates before irreversibly committing (Figure 2C), these observational studies do not exclude the possibility that gene expression noise is a byproduct of differentiation [23].

Functional roles of gene expression noise during lineage specification

Pioneering studies investigated the functionality of cellular noise by sorting mESCs based on the expression of fluorescent protein reporters for key regulators, followed by differentiation cues

[8,44]. NANOG-high, SOX2-high, and OCT4-high cells differentiate more efficiently into the neural, neuroectoderm, and neuroectoderm/mesoderm lineages, respectively, compared with their low-expression counterparts. These studies revealed that seemingly homogeneous cell cultures contain cells that are transiently in slightly different states, which can be biased in cell fate or predisposed for differentiation. To better reflect RNA variability, endogenous *Hhex* was tagged with a fluorescent reporter driven by a synthetic internal ribosomal entry site, resulting in high translational efficiency [9,45]. This study shows that *Hhex*-high mESCs are more primed for primitive endoderm differentiation when compared with *Hhex*-low or unsorted mESCs [9]. While these sorting studies effectively explore a functional role of gene expression noise, they are restricted to one or a few genes because of the required genetic modification.

Exciting recent work has, for the first time, enabled the modulation of transcriptional noise at a genome-wide level. Desai *et al*. screened for noise modulators that amplified RNA cell-to-cell variability without changing mean expression levels [46]. They identified the compound 5′-iodo-2′-deoxyuridine (IdU), which increases the amplitude of transcriptional bursts while shortening their duration, resulting in higher global transcriptome variability. Interestingly, IdU treatment enhanced both neuroectodermal differentiation and reprogramming efficiency, supporting the idea that elevated transcriptional noise may facilitate cellular state transitions. In a complementary approach, García-Blay *et al*. screened for gene expression noise regulatory proteins in mESCs [47], identifying SON, a nuclear speckle and splicing-associated protein. SON knockdown changes mRNA noise levels of genes associated with differentiation. Interestingly, this leads to a decreased ability of sorted NANOG-high and NANOG-low mESCs to reinstate the original heterogeneous mESC population, demonstrating that SON is important for cell-state switching.

Further *in vivo* evidence for a functional role of gene expression noise originates from the observed developmental fate biases between cells at the two-cell stage in humans [48] or the four-cell stage in mice [40,49]. Knocking down *Sox21* expression (which is highly variable at the four-cell stage) in a single mouse blastomere at the late two-cell stage results in fewer asymmetric cell divisions, leading to more trophectoderm and fewer ICM cells at the late blastocyst stage [40]. Exciting new findings at the 16-cell stage in mice suggest that not a single gene, but the stochastic activation of two or more genes, is required for lineage commitment towards the epiblast [50]. Using a *Nanog* KO-approach, the authors suggest that NANOG, subject to stochastic gene expression itself, stabilises the stochastic coexpression of other pluripotency factors, such as OCT4, SOX2, or KLF4, to activate the epiblast cell specification. Recently, by identifying genes that mirror *Nanog* expression variability combined with gene regulatory network inference analysis, *Sox2*, *Pecam1* and *Hnf4a* were characterised as potential *Nanog*-cooperating genes to establish the epiblast fate [51]. Together, these experiments strongly suggest that gene expression variability at these early developmental stages facilitates cell fate decisions (Figure 2C).

Deciphering gene expression noise

To understand how the various sources of gene expression noise are functionally relevant, a novel challenge arises: how to decode this noise? Dissecting gene expression noise into **transcriptional bursting** and post-transcriptional processes and untangling intrinsic and extrinsic factors are critical to shedding light on the sources of the observed RNA variability [34,35]. While targeted studies have been valuable in examining the role of gene expression noise for developmental or pluripotency regulators, genome-wide studies provide a comprehensive view, allowing unbiased insights into the full transcriptome. Conventional scRNA-seq profiling is powerful to characterise cell-to-cell variability and allows correlations with known sources of noise, such as cell cycle [52], but cannot discriminate extrinsic from intrinsic noise, nor is it designed to decipher RNA

transcription dynamics. Therefore, advanced sequencing-based methods beyond conventional scRNA-seq are essential (Figure 3, Key figure).

Dissecting intrinsic and extrinsic noise

Intrinsic and extrinsic noise have traditionally been assayed using two distinctly allelic fluorescent reporters controlled by two identical promoters [2]. Allele-specific scRNA-seq has recently allowed the extension of such measures to a transcriptome-wide scale (Figure 3). Of note, this powerful method depends on sufficient heterozygous SNPs for allelic calling and presents some (technical) limitations inherent to scRNA-seq (Box 2) [31,41,57,59,70–73]. Using mouse embryonic fibroblasts (MEFs) and mESCs, genes containing a TATA box consistently show higher intrinsic and extrinsic noise [31,72,73]. Unsurprisingly, cell cycle genes demonstrate high extrinsic but low intrinsic noise [10,72]. Also, genes encoding extracellular or secreted proteins exhibit high extrinsic and intrinsic noise, likely reflecting the lack of evolutionary pressure to reduce noise for these genes [72].

Within a developmental context, the use of mESCs shows that genes associated with development and cell identity exhibit elevated gene expression noise, while housekeeping genes show more stable expression patterns with low cell-to-cell variability [31,73]. Notably, developmental regulators and transcription factors, such as Nanog and Dnmt3b, exhibit high intrinsic but low extrinsic noise [10,31,73], suggesting that their variable expression is an inherent feature of mESCs. Altogether, these findings underscore that distinct noise patterns arise for different gene classes.

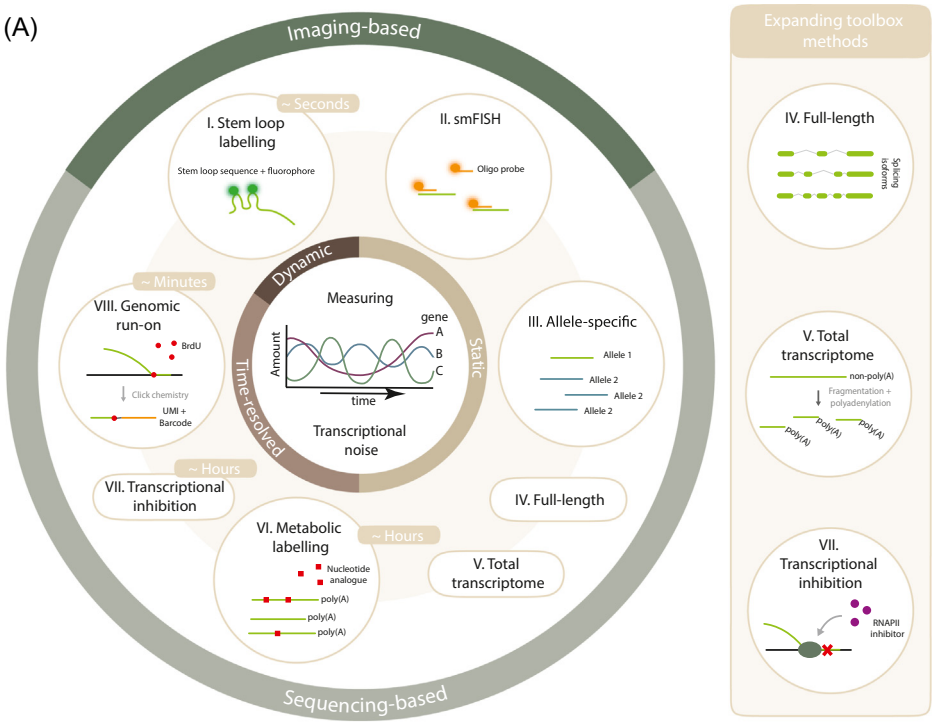
Transcriptional bursting

As genome-wide scRNA-seq has an inherent limitation in that it lacks temporality, its use for capturing transcriptional bursting dynamics requires additional procedures (such as modelling of transcriptional dynamics: Box 3). Alternatively, new omics approaches have been developed to comprehensively assay transcriptional bursting, such as global run-on followed by sequencing (GRO-seq) [82] (Figure 3). In the recently developed single-cell GRO-seq, single nuclei are sorted after run-on with a nucleotide analogue [62], the labelling of which allows for single-cell resolution of transcriptional activity upon sequencing. In mESCs, this revealed that developmental and cell identity genes undergo pronounced burst-like transcription. By inferring a temporal dimension based on the elongation speed of RNA polymerase, the authors show that RNA bursting at enhancers temporally precedes the transcriptional bursting of their target genes, exemplified for the pluripotency genes *Nanog* and *Sox2*. While scGRO-seq is powerful to estimate **transcriptional bursting parameters**, it does not provide information on the steady-state RNA pool, preventing integration with other contributors to gene expression noise, such as RNA degradation. Additionally, current scGRO-seq protocols are limited to *in vitro* applications.

A recent alternative method for estimating bursting parameters is represented by the integration of metabolic labelling in scRNA-seq workflows [27] (Figure 3). In this set-up, samples are incubated with a nucleotide analogue, which gets incorporated throughout the transcriptome and can be detected during subsequent sequencing. This approach, combined with statistical modelling, allows for the estimation of bursting parameters [70,79,83]. Additionally, this technique allows for correlating RNA bursting and degradation, revealing a negative correlation between burst frequency and RNA degradation rate for highly expressed genes in MEFs [70]. In human K562 cells, reports suggest either a positive [79] or negative [70] correlation between these parameters. Whether these differences are dependent on context (gene selection or experimental variation), method (modelling of transcriptional dynamics; Box 3), or other factors remains to be determined. Within a developmental context, it will be interesting to estimate similar kinetic parameters within mESCs, for which the data are available [84]. Of note, inherent to the scRNA-

Key figure

Overview of state-of-the-art techniques used to measure and decipher gene expression noise



(B)

	Multiplexing	Time resolution	Single cell resolution	Intrinsic & extrinsic deconvolution	Transcriptional bursting	RNA degradation	Technical limitations	In vivo limitations
Stem loops	<10 genes	~ Seconds	Yes	Yes*	Yes, direct	No	Spot detection	Optical transparency & phototoxicity
smFISH	>1000 genes	Static	Yes	Yes*	Estimation, modelling	No**	Spot detection	Fixation sample
Allele-specific	Genome-wide	Static	Yes	Yes	Estimation, modelling	Estimation, modelling	Allelic calling	Heterozygous SNPs
Full length	Genome-wide	Static	Yes	No	TBD	TBD	TBD	TBD
Total transcrip-tome	Genome-wide	Static	Yes	No	TBD	TBD	TBD	TBD
Metabolic labelling	Genome-wide	~ Hours	Yes	No	Estimation, modelling	Yes (pulse-chase)	Sparsity nascent reads & poly(A)	Injection nucleotide analogues
Transcript-ional inhibition	Genome-wide	~ Hours	Yes	No	Estimation, modelling	Yes	Affecting cellular processes	Affecting cellular processes
Genomic run-on	Genome-wide	~ Minutes	Yes	No	Yes, direct	No	Sparsity nascent reads	Single nuclei isolation

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(See figure legend at the bottom of the next page.)

Box 2. Tackling technical noise

Single-cell sequencing–based profiling methods are notoriously known for technical noise, which is caused by factors such as variability in capture efficiency, sampling noise, amplification biases, and dropout [54,55]. This complicates the dissection of technical and gene expression noise and can result in an overestimation of the biological noise when compared with imaging. Paradoxically, sequencing-based approaches may underestimate changes in noise upon perturbations, such as IdU treatment, potentially due to nonlinear data transformation [56]. Furthermore, low-expressed genes suffer disproportionately from dropout effects [57,58]. For various pluripotency genes, such as Nanog, which are lowly expressed and highly variable in mESCs, technical noise and dropout in scRNA-seq can obscure inherent cell-to-cell variability [56].

Good experimental and computational practices can mitigate some of these issues. First, adding synthetic RNA spike-ins [59,60] allows the estimation and correction of capture efficiency and technical variation, improving quantification accuracy, especially for low-expressed genes. Second, advanced normalisation [54,61] can reduce technical noise without distorting true biological variability. However, these methods may also filter out extrinsic noise, which can be relevant in developmental contexts. Third, if the goal is to focus exclusively on intrinsic noise, scRNA-seq allows, to some extent, to account for extrinsic noise, such as cell cycle phase or metabolic state [52]. Interestingly, cell cycle analysis improves when based on nascent transcriptome profiling [62].

Despite best practices in combination with deep sequencing, accurate quantification for low-abundance transcripts remains a major challenge [54,58]. This challenge becomes more apparent when shifting towards specialised single-cell approaches as other layers of complexity are added. Quantification bias for low-expressed genes has been demonstrated for metabolic labelling assays [63]. Additionally, undesired RNA degradation during sample processing can disproportionately affect unstable RNA [64]. Inaccurate quantifications can propagate to invalid estimations of bursting, as has been reported for Nanog [31]. Recent efforts to improve capturing efficiency and reduce cell loss could improve these quantification biases for low-expressed genes such as transcription factors [65]. Furthermore, simulation-based approaches can be used to determine the effect of technical noise in a controlled manner [59,60].

Arguably, smFISH and its multiplexed derivatives (MERFISH, seqFISH+) allow the most accurate quantification of gene expression noise because of limited technical noise [66–69]. For this reason, these methods are often used as validation tools for gene expression noise estimations from sequencing-based assays [46,56].

seq metabolic labelling approach, in which labelling is usually performed for 1 h or longer, the data are static, lacking temporal resolution for exact quantification of transcriptional bursting parameters.

To include temporal resolution, real-time readouts of RNA levels through live imaging assays provide the most accurate measures, such as stem-loop labelling [85,86] (Figure 3). However, these assays are generally low-throughput and often require laborious genomic integrations. Also, simultaneously monitoring the bursting of several genes is challenging because of the limited stem-loop integrations that are available (mainly MS2 and PP7). As such, less-invasive imaging with none or limited genomic editing can be used, such as live single-molecule fluorescence *in situ* hybridisation (smFISH) [87,88]. Furthermore, advanced smFISH methods allow validation of quantifications of nascent and total RNA (Box 2) [66], but these assays are challenging to set up.

RNA degradation

Following RNA synthesis, the rate of degradation contributes to gene expression noise. Genome-wide **RNA turnover** can be assayed through metabolic labelling. Specifically, pulse-chase experiments allow accurate estimations of **RNA stability** by determining **RNA half-lives** [27,84]. Their single-cell resolution further allows RNA stability analysis of heterogeneous populations, important for determining gene expression noise [84,89,90]. In mESCs, this allowed identification of genes associated with either low stability (mainly genes involved in transcriptional regulation

Figure 3. (A) Schematic overview of imaging-based (I,II) and sequencing-based techniques (III–VIII), classified by measurements being static (II–V), time-resolved (VI–VIII), or dynamic (I). (B) Table that summarises the key characteristics, limitations, and applications of each of the methods presented in the overview. *Intrinsic and extrinsic discrimination is possible when integrating different stem-loop sequences per allele or using allele-specific probes for stem-loops and smFISH, respectively. **While not common, there have been approaches that estimate RNA degradation using smFISH data [53]. TBD, to be determined; smFISH, single-molecule fluorescence *in situ hybridisation*

Box 3. Modelling transcriptional dynamics

Quantitative descriptions of the transcriptional dynamics require the assumption of an underlying mechanistic model of transcription. A basic mechanistic model defines two states, an ON state and an OFF state [74]. A gene in the ON state is actively transcribed by the transcription machinery, when compared with the OFF state, in which the gene is transcriptionally inactive. In reality, transcription is a less binary and rigid process in view of the steps involved, including the binding of transcription factors and RNA polymerase II [6]. As a result, several studies have adopted a multistate model, such as a three-state model [26,75,76], or a continuum model [77] of transcriptional dynamics (Figure I).

Recent reviews thoroughly examine these various models, present mathematical representations of key parameters such as burst size and burst frequency, and describe algorithms for parameter inference [26,78]. While multistate models might reflect bursting more accurately, parameter inference for all state switching rates requires high-resolution and rich datasets, which are not available for most sequencing-based assays due to the inherent sparsity of these methods. Therefore, most inference models assume a two-state telegraph model [25,70,73,79]. Nonetheless, even this oversimplified model requires sufficient data, where reports can range from at least several hundred to more than 1000 cells with 10 000 to 50 000 unique reads per cell [70,73]. As direct comparison with imaging data is often lacking, it is unclear to what degree fitting with the simplistic two-state model affects bursting parameters, both quantitatively and qualitatively. Of note, gene expression noise is more pronounced for genes associated with pluripotency and development, representing highly dynamic transcriptional modes. Therefore, oversimplifying their dynamics may make characterization of their bursting mechanisms especially challenging.

Besides several parameter inference methods based on statistical methods, deep learning–based inference has been proposed for modelling transcriptional dynamics [80,81]. These models hold the promise to capture more complex and multistate transcriptional dynamics using scRNA-seq data, while handling high-dimensional and sparse datasets [80]. However, deep learning methods have the inherent limitation of being less interpretable [81] and require large, high-quality training data [80].

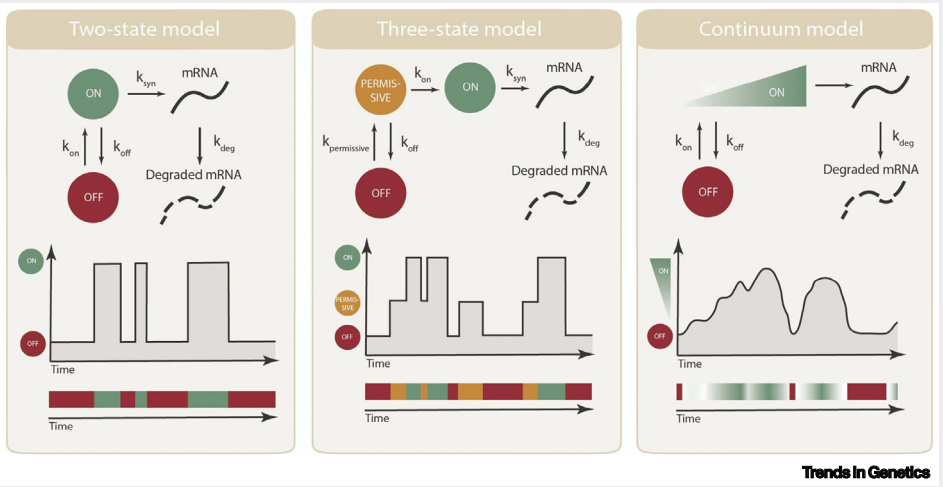

Figure I. Schematic representation of common models of transcriptional dynamics. From left to right: two-state telegraph model, three-state model, and continuum model. The upper panels represent a schematic of the states and products with the associated state-switching or production/degradation rates. The lower panels represent the promoter state over time.

and developmental processes) or high stability (mainly genes involved in physiological processes and housekeeping genes) [27,84,89]. Interestingly, these studies suggest that the unstable developmental and pluripotency genes have higher gene expression noise, partly due to rapid RNA degradation, whereas housekeeping genes are highly stable, showing low noise.

Expanding the gene expression noise toolbox

Next to transcriptional bursting and RNA degradation, RNA turnover is tightly interlinked with post-transcriptional processes such as splicing dynamics and nuclear export [6]. Full-length

single-cell RNA profiling methods such as Smart-seq3/Smart-seq-total [91], Nano-SINC-seq [92], and FLASH-seq [93] allow probing these post-transcriptional processes (Figure 3), of which splicing is considered to be a critical source of gene expression noise [6]. Recently, full-length profiling has been combined with a metabolic labelling approach, revealing that cell-to-cell RNA variability is also evident in intergenic regions and increases before decision-making [94]. The amount of nonspliced RNA can additionally serve as a proxy for nascent mRNA levels [95], potentially useful to estimate bursting dynamics. However, many transcription factors are intronless or yield a low abundance of unspliced reads, making splicing-based inference of bursting dynamics for these genes challenging [96].

Many scRNA-seq approaches are based on poly(A)-tail capture. As a result, not all transcription initiation events will be recorded [97], potentially leading to an underestimation of transcriptional activity [62]. Also, it does not allow the assay of nonpolyadenylated RNAs, such as noncoding (nc)RNAs, which have recently been shown to demonstrate stochastic expression as well [94]. Stochastic transcription of ncRNAs can affect essential cellular processes, as shown for Xist and Tsix, involved in X-chromosome inactivation [98]. Several novel total transcriptome profiling methods, including VASA-seq [99] and NERD-seq [100], aim to detect nonpolyadenylated RNAs and thus might prove useful for refining quantifications (Figure 3).

Additionally, more direct measurements of RNA degradation may serve as a valuable complementary approach to metabolic labelling (Figure 3). The block of transcription, using RNAPII inhibitors such as α-amanitin or actinomycin D, followed by scRNA-seq to probe for RNA dynamics over time, allows validation or refinement of half-life estimates [101]. Compared with metabolic labelling experiments, these assays do not suffer from labelling bias [63]. However, these approaches might induce stress and thereby unintentionally change degradation rates. An alternative method that does not disrupt cell physiology involves a gene-specific block of transcription using controllable synthetic promoters [102].

Insights into cobursting and allelic coordination of bursting

The recent possibilities to assay bursting kinetics at a genome-wide level have highlighted a phenomenon where the bursting of two genes is synchronised, called **cobursting** or coordinated bursting, as first identified in nonmammalian systems through live-cell imaging (Box 1). A major challenge is that most sequencing-based assays do not reach the time resolution to confidently call for cobursting. In such cases, we will refer to **correlated transcription**. Recent genome-wide analysis into correlated transcription in mESCs using scGRO-seq shows that it is rare, occurring in 0.7% of all gene pairs [62]. Correlated transcription (at 4-min time resolution) is predominantly among genes involved in cell cycle, splicing, and circadian rhythm. In line with this, allele-specific scRNA-seq, combined with metabolic labelling, identified only a few gene pairs demonstrating significant correlated transcription in mouse fibroblasts [70]. These pairs were mainly paralogues and/or genes in close spatial proximity. This agrees with recent findings that transcriptional correlations between genes occur almost exclusively in *cis*, while transcriptional bursting remains largely independent [103].

These allele-specific assays have also revealed that there can be unequal expression and thus differences in the bursting parameters of the two alleles in the same cell [57,71]. The degree of allelic coordination of the bursting kinetics differs amongst genes. During reprogramming from MEFs to induced pluripotent stem cells (iPSCs) [71] and in mouse blastocyst cells [57], many genes, such as those involved in ribosome biogenesis, protein folding, and stem cell maintenance, demonstrate the same bursting kinetics for both alleles [71]. In pregastrulation embryos, developmental genes, such as *Brachyury T*, can demonstrate coordinated allelic bursting patterns of their alleles

[104]. Whether bursting is coordinated depends on chromatin accessibility, histone marks, and differential transcription factor binding [71]. At the extreme, independent allelic bursting can give rise to a phenomenon called dynamic autosomal random monoallelic expression (aRME) [104,105]. Dynamic aRME has been suggested to enhance cell-to-cell variability, which leads to cellular fate priming during lineage commitment in pregastrulation embryos [104]. When aRME is retained in future generations of the clonal population, clonal RME can arise [105]. Clonal RME has been shown to be enhanced upon differentiation, although it is not fully understood yet [105,106].

Overarching themes of noise in mammalian development

Given the measured increase in gene expression noise at the onset of differentiation (Figure 2), a key question is how this contributes to the lineage commitment of individual cells. Several genetic modelling studies have investigated the role of noise during cell specification [107–109], such as ICM differentiation [107]. Here, extrinsic noise is modelled by randomly varying the concentration of extracellular Fgf4 perceived by each cell, while intrinsic noise is modelled by stochastic fluctuations of Nanog, Gata6, FGFR, Erk signalling, and secreted Fgf4, key factors involved in the regulatory network [107]. The model suggests that the presence of extrinsic noise seems to better represent *in vivo* observations of cell specifications than intrinsic noise. A recent study modelled ICM differentiation using an artificial intelligence (AI)–assisted parameter inference scheme to determine which core regulatory and signalling parameters best recapitulate *in vivo* observations, incorporating various sources of noise [110]. Interestingly, recent live-cell imaging in mouse embryos during ICM differentiation indicates that a stochastic burst of NANOG at the 32-cell stage is correlated with primitive endoderm-primed cells switching towards the epiblast fate [111]. In *in vitro* fibroblasts, heterogeneity in response to signalling has been shown to be dependent on cell-to-cell variability [112,113]. To what degree gene expression noise also influences the response to instructive signalling in a developmental context remains yet to be determined. However, it seems that cell-to-cell variability can increase the amount of transmitted information within cellular responses [114]. New computational models have emerged to focus on the dynamics of fate transition [96,115]. These models indicate that cells make fate choices by either a deterministic one-or-another decision regulated by signalling or a stochastic split between two fates. For the latter, the balance between the fates is influenced by noise. Interestingly, also when modelling these transitions, adding noise '*in silico*' is beneficial to drive cells out of the progenitor state towards differentiation [115].

Recent advanced single-cell methods have started to characterise the genes showing high levels of expression noise. In these assays, mESCs show enhanced transcriptional noise specifically in developmental and pluripotency genes [31,73], likely due to lower transcriptional burst frequency and larger burst size combined with reduced RNA stability [27,84,89]. Given the particularly enhanced gene expression noise for developmental regulators, the use of targeted approaches to further investigate the noise dynamics of these genes might be useful.

The link between transcriptional noise and specific bursting parameters, combined with the observed enhanced noise in specific gene classes, sparks questions about noise regulation. How epigenetic factors contribute to transcriptional dynamics has been reviewed before [78,116]. Combined with recent genome-wide single-cell assays [31,72,83], these studies highlight the correlation between TATA box promoters and genes exhibiting higher transcriptional noise generated by increased burst sizes. Genes with TATA box promoters are typically associated with dynamic regulatory functions such as development [25,73], whereas housekeeping genes tend to contain CpG-island promoters [116], for which unmethylated CpG islands have been

associated with reduced transcriptional noise [83,116]. Of note, positive correlations between burst frequency and H3K27ac [70,73,79] and between H3K4me3 and burst size [70,79] have been observed. Finally, the binding of Polycomb Repressive Complex 2 (PRC2), enriched at promoters of developmental genes, is linked to higher transcriptional noise [31]. With the emergence of multimodal single-cell epigenetic readouts for features such as chromatin accessibility and DNA methylation, combined with a transcriptome readout, new avenues of studying cell-to-cell variability in epigenetics and coupling this to transcriptional noise may arise. Also, recent time-resolved lineage tracing [117,118] will allow us to trace back the epigenetic state around cell fate decisions, shedding light on how cells make decisions and, perhaps, linking this back to the role of gene expression noise.

Concluding remarks and future perspectives

The recent advances into measuring and manipulating gene expression noise have resolved important outstanding questions concerning the presence and function of noise, mainly in relation to its contribution to cell fate decisions. Upcoming single-cell profiling technologies now allow the estimation of transcriptional bursting and RNA degradation parameters, in addition to many extrinsic sources of noise. The integration of such methods will be even more powerful in decoding the mechanisms behind gene expression noise. A recent pioneering study combining allele specificity with metabolic labelling enabled the quantification of transcriptional bursting and RNA degradation with allele specificity [70]. Together with future studies, this will shed light on the coordination of allelic expression or coburstin between genes, a key insight for understanding how epigenetic regulation shapes transcriptional noise (see Outstanding questions). By being able to dissect, quantify, and, perhaps, predict noise at the single-cell level, we envision that the term 'noise' will soon not be fitting anymore for a range of features that are currently defined as noise, especially for the extrinsic contributors [119].

So far, the functional role of gene expression variability in mammalian development has mostly been studied using mESCs. Typically, this system does not reflect features of *in vivo* development such as cell type diversity, extensive cell–cell communications, and signalling gradients. Therefore, an attractive next step will be to apply the advanced transcriptomics technologies, as discussed in this review, to understand gene expression noise in an *in vivo* setting. While largely feasible in mice, this is challenging or impossible in humans because of technical and ethical reasons and the limited availability of IVF embryos. However, new embryo-like models such as blastoids, gastruloids, and somitoids are promising alternatives [120]. Applying advanced single-cell transcriptomic methods on these models creates the possibility to study and manipulate RNA variability and its regulators in a context that closely resembles early human embryogenesis, showing symmetry breaking and patterning [120]. This would deepen our understanding of gene expression noise dynamics during embryogenesis and its role in guiding lineage specification.

Outstanding questions

Can we identify additional controlled strategies to perturb bursting at a genome-wide scale to enable further functional studies?

What is the functional role of increased gene expression noise during lineage transitions?

How are the changing cell-to-cell RNA variability levels during embryonic development regulated?

How conserved are recent findings in mouse embryonic stem cells on gene expression noise in human embryonic stem cells, and to what extent do these findings reflect the *in vivo* situation in embryos?

Is there also noise at the epigenetic level, and if so, to what extent are transcriptional noise and epigenetic noise related?

How do bursting parameters, as determined by static sequencing approaches, relate to bursting parameters obtained from dynamic microscopy approaches, such as live-cell imaging, and how to reconcile potential differences?

How widespread is coordinated allelic bursting, and what are the specific mechanisms that regulate coordinated allelic bursting?

Why does coburstin occur only for a small subset of gene pairs with similar regulatory mechanisms, and is there a specific mechanism that regulates the coburstin for these genes?

Would we be able to predict a developmental lineage bias of individual cells based on the quantification of individual contributors to gene expression noise?

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Declaration of interests

The authors declare no competing interests.

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